

Unit 03: Disk Quotas, Storage Management, and Network Configuration

3.1 Disk Quotas and Storage Management

Disk Quotas

Disk quotas are limits imposed on the amount of disk space and number of files (inodes) that a user or group can use on a filesystem. They help prevent any single user from consuming all disk resources, ensuring fair usage and system stability.

- **Purpose:** Prevent a single user from consuming all disk resources.

Commands and Steps:

1. Enable quotas on a filesystem:

```
sudo mount -o remount,usrquota,grpquota /home
sudo quotacheck -cum /home      # Check quotas
sudo quotaon /home              # Enable quotas
```

2. Set user quotas:

```
sudo edquota username
```

- Define **soft limit** (warning threshold) and **hard limit** (maximum allowed).

3. View quotas:

```
quota -u username      # User quota
repquota /home          # All users on /home
```

Disk and Inode Limits:

- Disk limit: Maximum storage in bytes.
- Inode limit: Maximum number of files/directories.

RAID Implementation

RAID (Redundant Array of Independent Disks) is a method of combining multiple physical disks into a single logical unit to achieve data redundancy, improved performance, or both. RAID protects against disk failures and can enhance read/write speed depending on the RAID level used.

Common RAID Levels:

RAID Level	Description	Advantage	Disadvantage
RAID 0	Striping	High performance	No redundancy
RAID 1	Mirroring	Data redundancy	Uses 2x disk space
RAID 5	Striping + Parity	Balance of performance & redundancy	Minimum 3 disks, slower writes
RAID 6	Striping + Double Parity	Can tolerate 2 disk failures	Slower writes, min 4 disks
RAID 10	RAID 1 + 0 (Mirror + Stripe)	High performance & redundancy	High disk requirement

Software RAID commands (mdadm):

```
sudo mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 /dev/sda  
/dev/sdb  
sudo mdadm --detail /dev/md0          # Check RAID status
```

Logical Volumes (LVM)

Logical Volume Management (LVM) is a system that allows flexible partitioning, resizing, and management of storage. Unlike traditional disk partitions, LVM abstracts physical storage into logical volumes, making it easier to resize, expand, or combine storage without downtime.

Components:

1. **Physical Volume (PV):** Physical disk or partition.
2. **Volume Group (VG):** Pool of PVs.
3. **Logical Volume (LV):** Partition created from VG.

Commands:

```
sudo pvcreate /dev/sda  
sudo vgcreate vg_data /dev/sda  
sudo lvcreate -L 10G -n lv_storage vg_data  
sudo mkfs.ext4 /dev/vg_data/lv_storage  
sudo mount /dev/vg_data/lv_storage /mnt/storage
```

Advantages:

- Resize volumes dynamically (lvextend, lvreduce).

- Snapshots for backups.
- Combine multiple disks into a single volume.

3.2 Network Configuration

IPv4 and IPv6 Addressing

- **IPv4:** 32-bit address (e.g., 192.168.1.10)
- **IPv6:** 128-bit address (e.g., 2001:0db8:85a3::8a2e:0370:7334)

Configure IP Address (Temporary):

```
sudo ip addr add 192.168.1.10/24 dev eth0 # IPv4
```

```
sudo ip addr add 2001:db8::1/64 dev eth0 # IPv6
```

Configure IP Address (Permanent):

- Modify `/etc/network/interfaces` (Debian/Ubuntu) or `/etc/sysconfig/network-scripts/ifcfg-eth0` (RHEL/CentOS).

Check Configuration:

```
ip addr show
```

```
ip route show
```

```
ping 8.8.8.8 # Test IPv4 connectivity
```

```
ping6 google.com # Test IPv6 connectivity
```

Routing

Routing is the process by which a computer determines the path packets take to reach a destination network. Routers and hosts use routing tables to decide where to forward packets based on IP addresses.

- **View routing table:**

```
ip route show
```

- **Add route:**

```
sudo ip route add 192.168.2.0/24 via 192.168.1.1
```

- **Delete route:**

```
sudo ip route del 192.168.2.0/24
```

Network Troubleshooting Tools

Tool	Purpose
ping	Test connectivity to a host
tracert / tracepath	Trace path packets take to a host
netstat / ss	View open ports, active connections, and routing info
ifconfig / ip addr	Check interface configuration
dig / nslookup	Test DNS resolution
tcpdump	Capture network packets for analysis